

The empirical recurrence for OEIS A259522: recovering a conjecture that is not written down

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Abstract

OEIS A259522 records “Empirical recurrence of order 71”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 71$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 71, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 71 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A259522 is “Number of $(n+1) \times (6+1)$ 0..1 arrays with each 2×2 subblock having clockwise pattern 0000 0001 0011 0101 or 1111.”. It has offset 1 and begins

3816, 180370, 7802386, 348802846, 15503669216, 690701197282, 30760063948568, 1370113260798484, ...

The entry states:

Empirical recurrence of order 71 (see link above).

As of the “Last modified” line on the live entry (Jan 15 2017 13:12:00, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 129$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 71$ of the 129, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 71$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 142$ exact terms of the model returns order 71 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{71} c_i a(n-i),$$

c_1	92	c_{19}	−34066153397872597	c_{37}	−4531293422582507040913	c_{55}	67579269986
c_2	−2736	c_{20}	461887440890464142	c_{38}	951850487583940141365	c_{56}	41510232494
c_3	23056	c_{21}	−195458321599322920	c_{39}	4343279742915065989399	c_{57}	−254009575
c_4	394432	c_{22}	−3137734196121586607	c_{40}	−886077634841998848371	c_{58}	−2162575799
c_5	−9504527	c_{23}	3744045517078715763	c_{41}	−3273576213865400803688	c_{59}	−183117424
c_6	41492789	c_{24}	14988944430248699133	c_{42}	529250979558783830442	c_{60}	5714936420
c_7	641712373	c_{25}	−27985837299858430976	c_{43}	1923750257148696136586	c_{61}	962227059
c_8	−7817226277	c_{26}	−51037052721771344512	c_{44}	−180922326757911390724	c_{62}	−44548395
c_9	7643527182	c_{27}	134827724076343365456	c_{45}	−867145328065962654956	c_{63}	−18973301
c_{10}	331905040502	c_{28}	121964514389431792345	c_{46}	10512439369491082902	c_{64}	−8015031
c_{11}	−1665540204785	c_{29}	−466020114042089544203	c_{47}	292279245609922542794	c_{65}	1177648
c_{12}	−4553793882054	c_{30}	−190003947513410566188	c_{48}	21916514739247058113	c_{66}	1238796
c_{13}	57265161480210	c_{31}	1207893576954726173694	c_{49}	−7104057268077710384	c_{67}	171834
c_{14}	−48070448318043	c_{32}	134229280624908811435	c_{50}	−11546033870158570791	c_{68}	−27364
c_{15}	−967620086305900	c_{33}	−2403700624167057525305	c_{51}	11823112451351740570	c_{69}	−1515
c_{16}	2667505173670135	c_{34}	164852181942570360967	c_{52}	2996508331554060186	c_{70}	−298
c_{17}	8917463593944229	c_{35}	3724558381607843579897	c_{53}	−1239565669371861528	c_{71}	−2
c_{18}	−45638035696302222	c_{36}	−629838679043441851396	c_{54}	−456664427830584147		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 71, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $71 + 4$ terms removed returns order 71 as well, so $\deg q_{\min} = 71 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 13 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $71 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 71 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A259522.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.